
Pretrained Models Transfer Learning Techniques and Popular CNN Architectures

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Topics

Pretrained Models, Transfer learning techniques and popular CNN architectures:

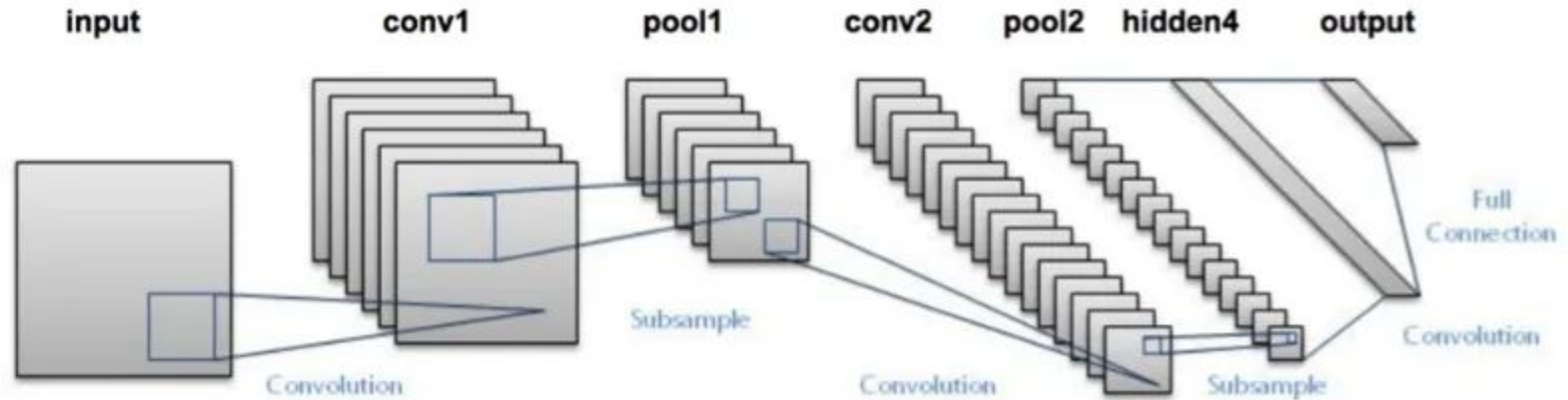
- ImageNet
- AlexNet,
- VGGNet
- ResNet
- InceptionV3
- DenseNet
- EfficientNet
- DarkNet
- TFLite

CNN

A Convolutional Neural Network (CNN, or ConvNet) are a special kind of multi-layer neural networks, designed to recognize visual patterns directly from pixel images with minimal preprocessing.

LeNet-5 (1998)

- LeNet-5, a pioneering 7-level convolutional network by LeCun et al in 1998, that classifies digits
- It was applied by several banks to recognise handwritten numbers on cheques digitized in 32x32 pixel grayscale input images.
- The ability to process higher resolution images requires larger and more convolutional layers, so this technique is constrained by the availability of computing resources.



ImageNet

The **ImageNet** project is a large visual database designed for use in visual object recognition software research.

The ImageNet project runs an annual software contest, the **ImageNet Large Scale Visual Recognition Challenge** ([ILSVRC](#)), where software programs compete to correctly classify and detect objects and scenes.

ImageNet Competition

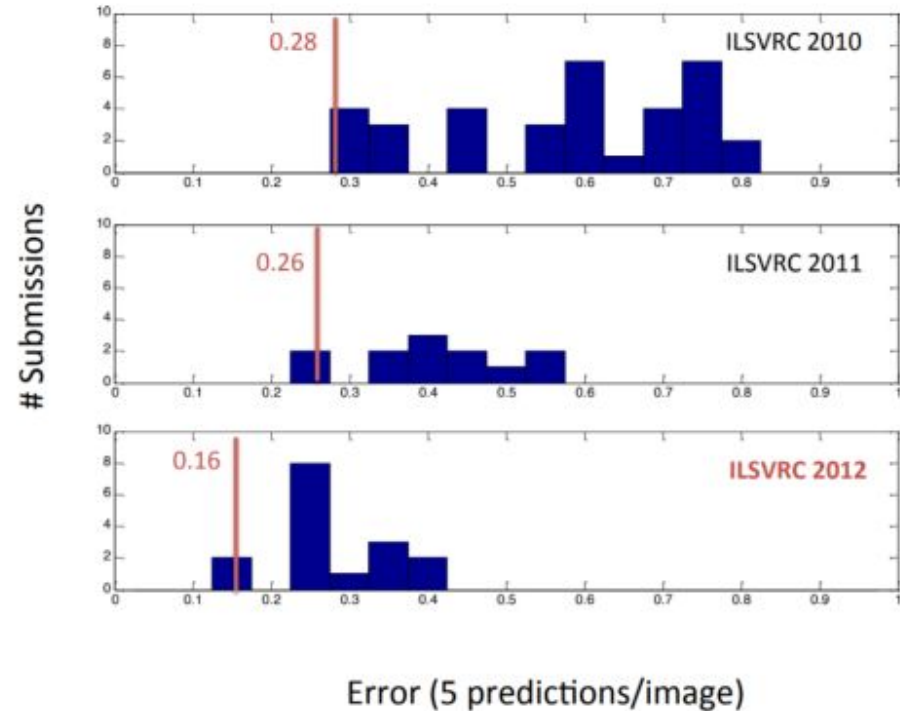
<https://image-net.org/challenges/LSVRC/2012/>

ImageNet Database

<https://www.image-net.org/>

Why AlexNet ?

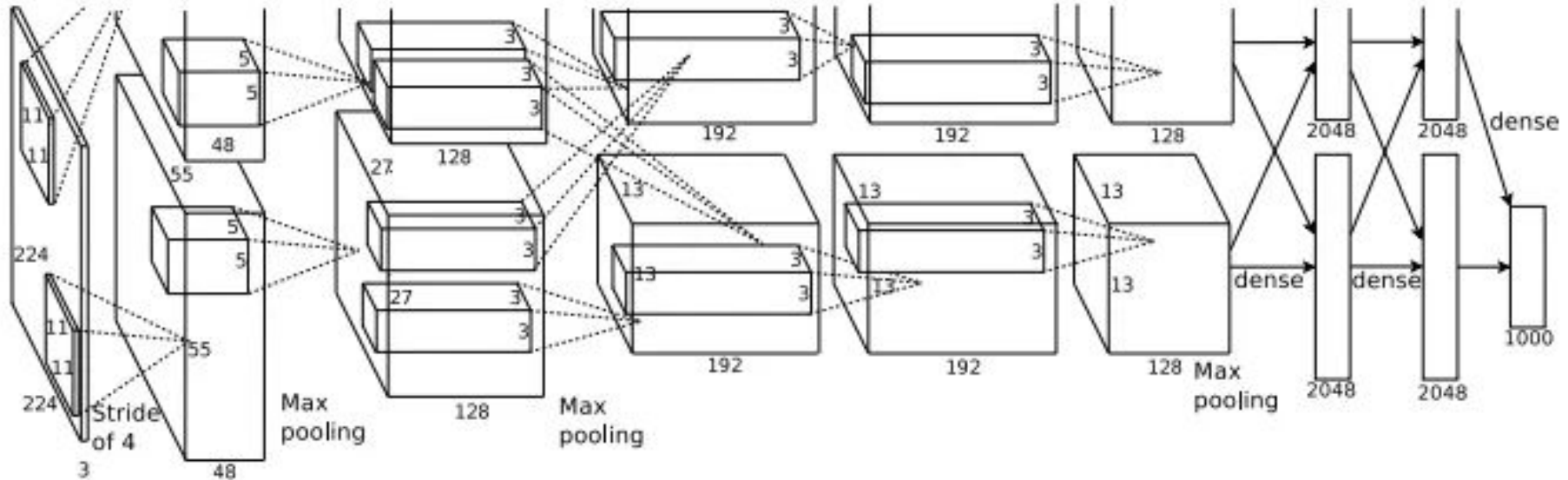
- AlexNet was born out of the need to improve the results of the ImageNet challenge.
- This was one of the first Deep convolutional networks to achieve considerable accuracy on the 2012 ImageNet ILSVRC-2012 challenge with an accuracy of 84.7% as compared to the second-best with an accuracy of 73.8%.



AlexNet (2012)

In 2012, [AlexNet](#) significantly outperformed all the prior competitors and won the challenge by reducing the top-5 error from 26% to 15.3%.

The second place top-5 error rate, which was not a CNN variation, was around 26.2%.



cont...

- The network had a very similar architecture as LeNet by Yann LeCun et al but was deeper, with more filters per layer, and with stacked convolutional layers.
- The network consists of 5 Convolutional (CONV) layers and 3 Fully Connected (FC) layers. The activation used is the Rectified Linear Unit (ReLU).
- 1000 different classes (e.g. cats, dogs etc.)
- It attached ReLU activations after every convolutional and fully-connected layer.
- AlexNet was trained for 6 days simultaneously on two Nvidia Geforce GTX 580 GPUs, (90 epochs) which is the reason for why their network is split into two pipelines.
- Image augmentation (flip, color changes, ..)
- Dropout
- AlexNet was designed by the SuperVision group, consisting of Alex Krizhevsky, Geoffrey Hinton, and Ilya Sutskever.

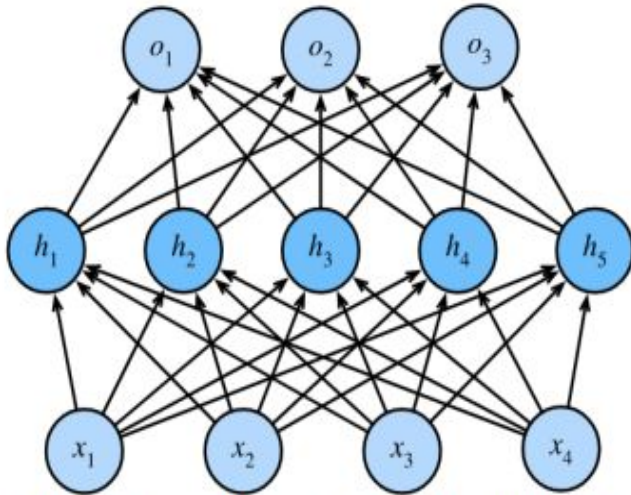
Ref: <https://learnopencv.com/understanding-alexnet/>

Ref: Krizhevsky, Alex & Sutskever, Ilya & Hinton, Geoffrey. (2012). ImageNet Classification with Deep Convolutional Neural Networks. Neural Information Processing Systems. 25. 10.1145/3065386.

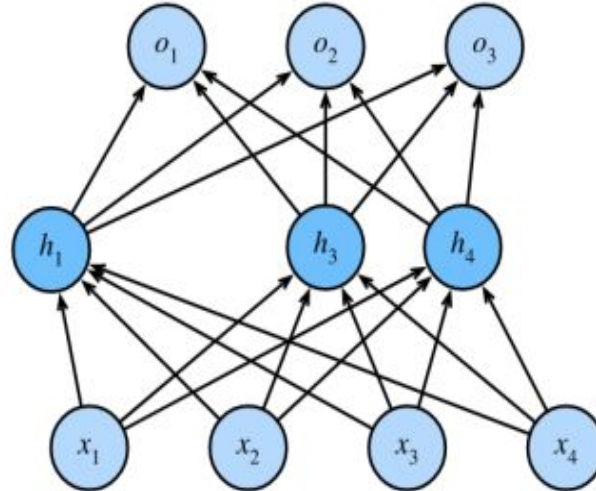
DROPOUT

- in some situations, the model is larger than we need – the model can be modified manually or using dropout technique
- **drop out** some neurons during training to avoid overfitting
- randomly **dropping out neurons**
 - neuron is dropped from the network with a probability of 0.5

Before dropout



After dropout



Data Augmentation



Mirror Image



Random Crops

227

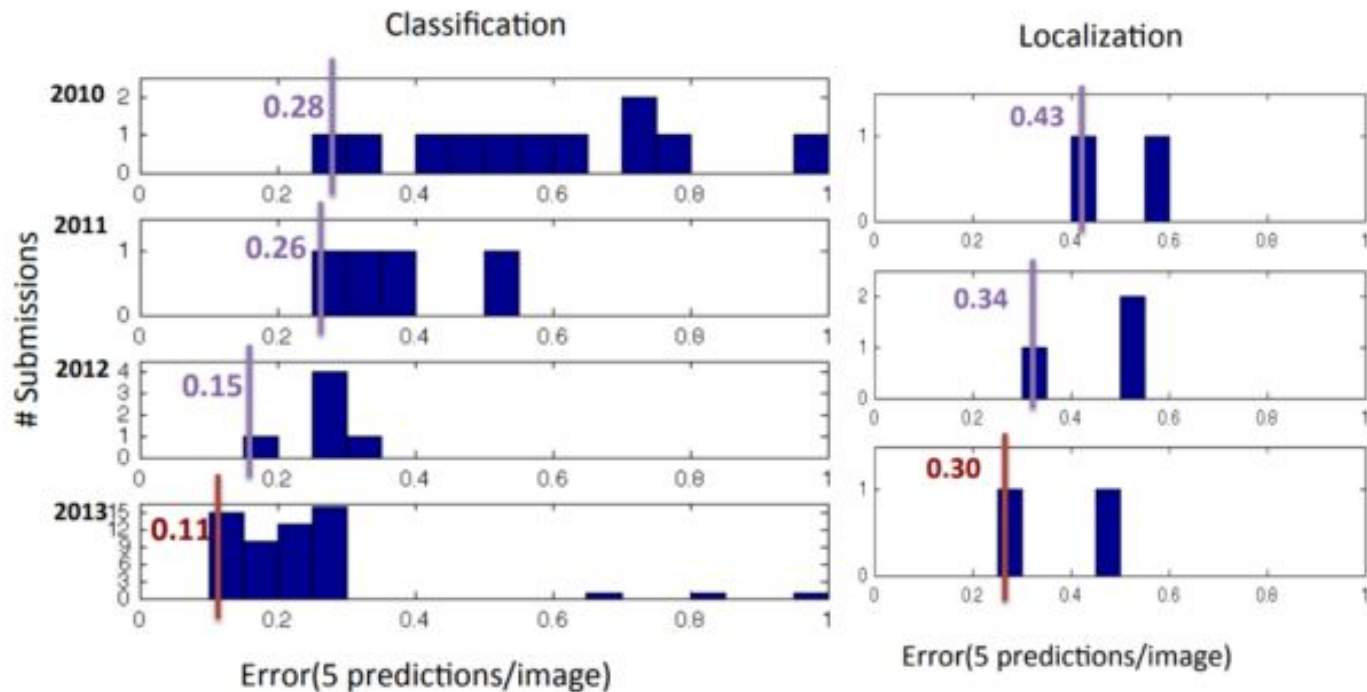


227

256

2013

ILSVRC over the years

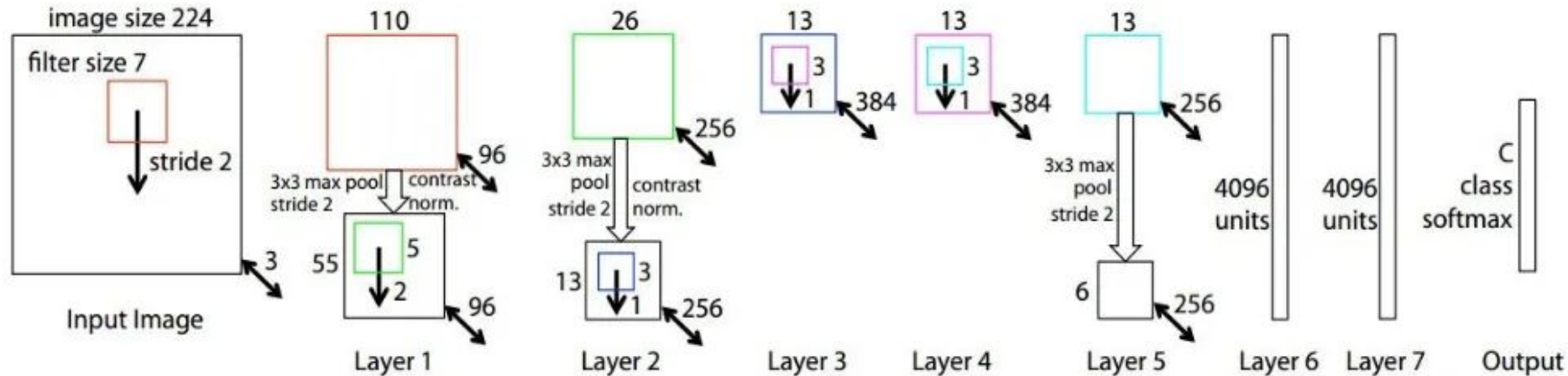


<http://image-net.org/challenges/LSVRC/2013/>

http://www.image-net.org/challenges/LSVRC/2013/slides/ILSVRC2013_12_7_13_clsloc.pdf

ZFNet (2013)

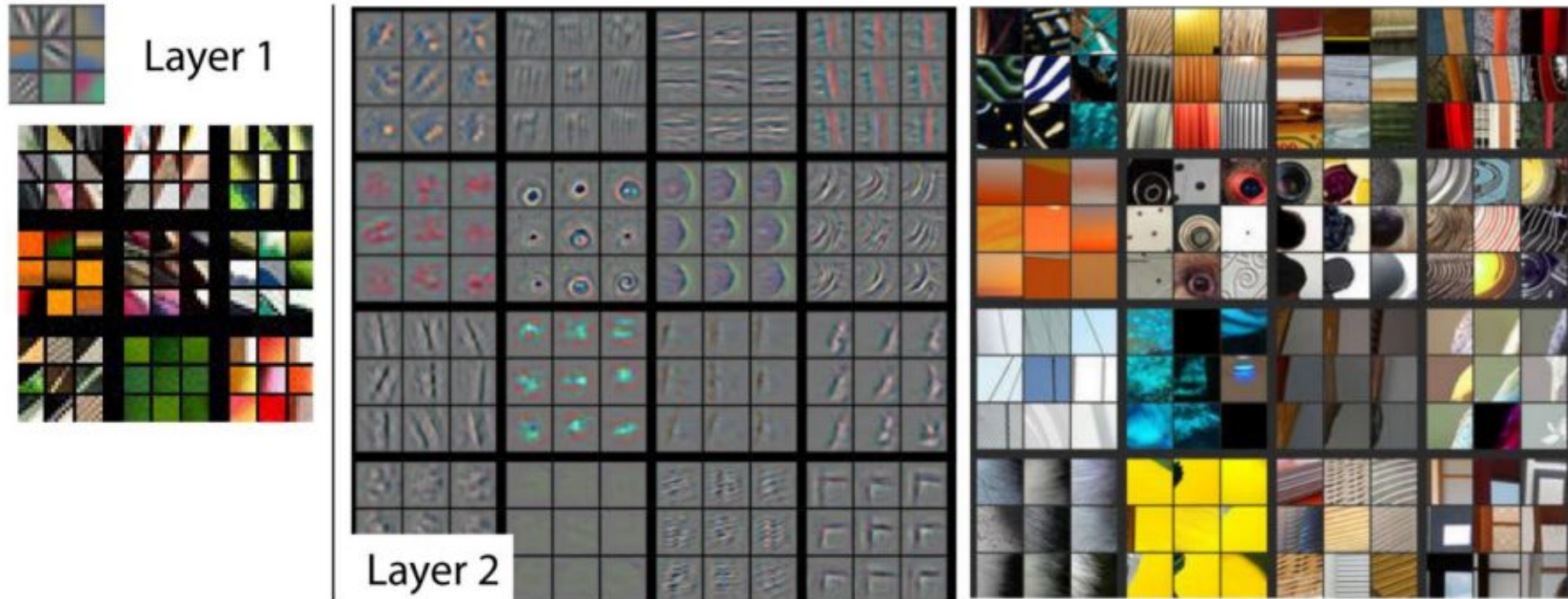
- The ILSVRC 2013 winner was also a CNN which became known as ZFNet.
- It achieved a top-5 error rate of 14.8% which is now already half of the prior mentioned non-neural error rate.
- It was mostly an achievement by tweaking the hyper-parameters of AlexNet while maintaining the same structure with additional Deep Learning elements.



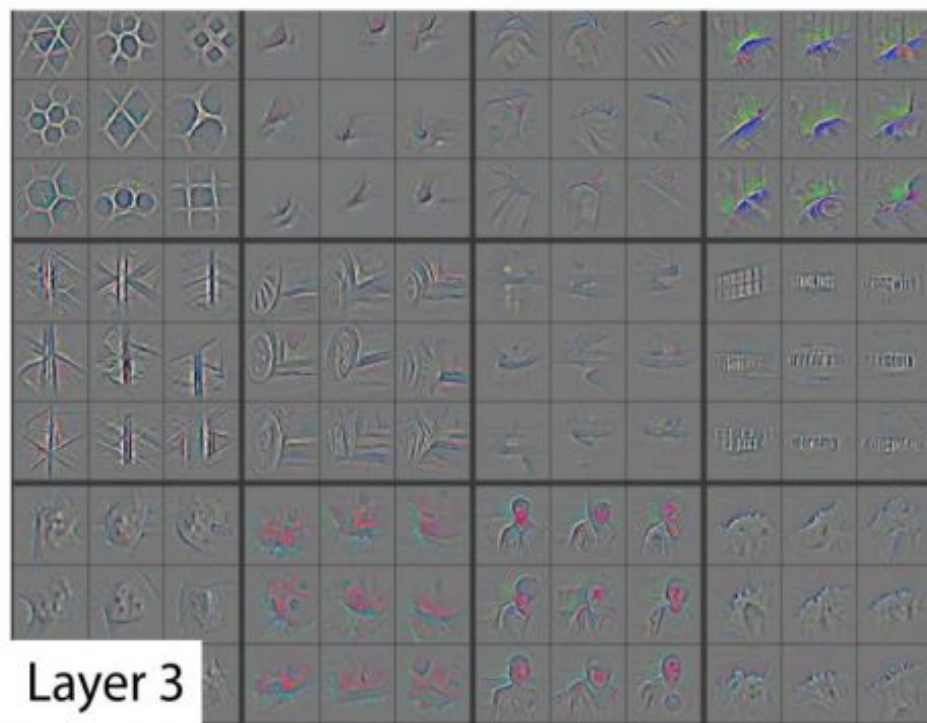
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- Similar architecture to AlexNet
- 11x11 vs. 7x7 filters in the first layer
- Number of filters is increasing
- Visualization of features - Deconvolutional Network

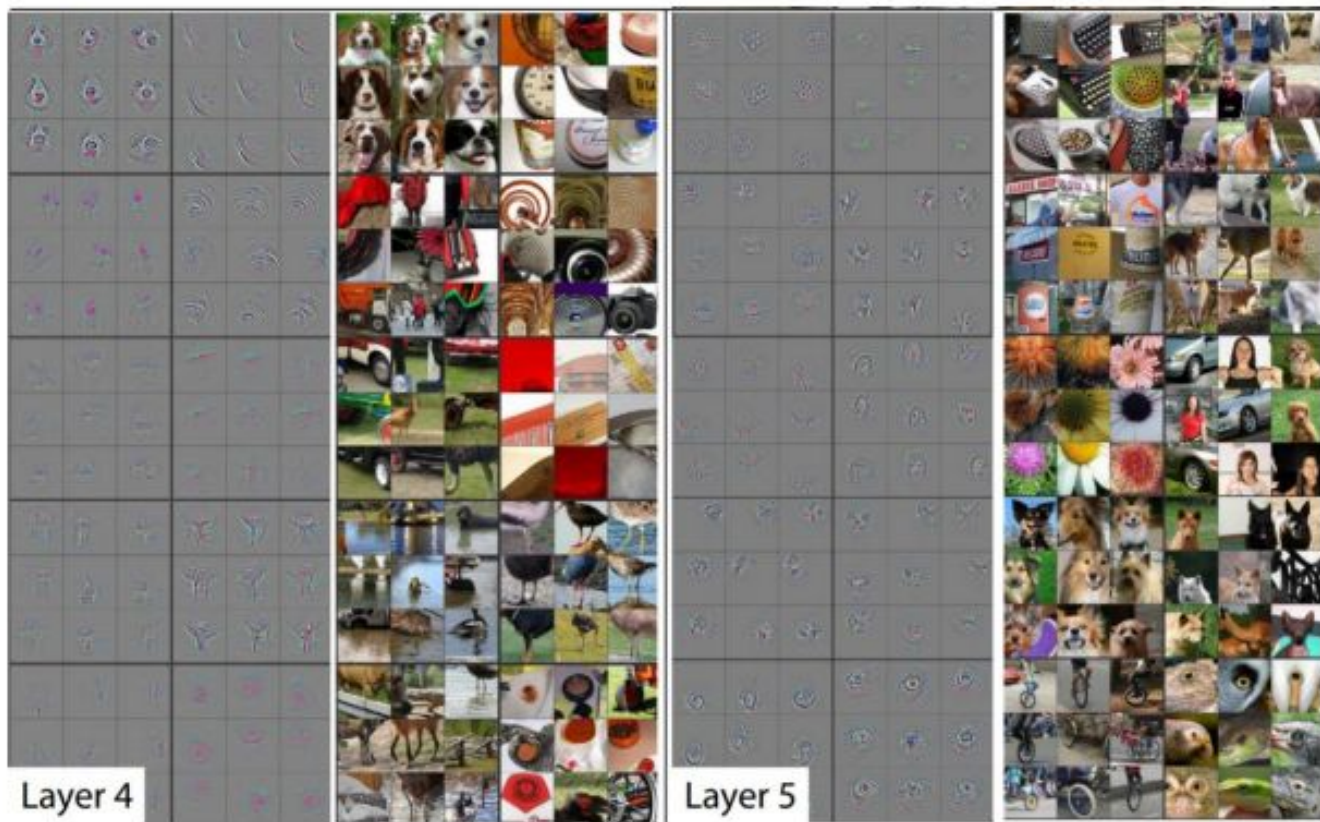
First Layers - low level feature detector (edge detector, circular features)



Deep Layers - higher level features such as dogs faces or flowers



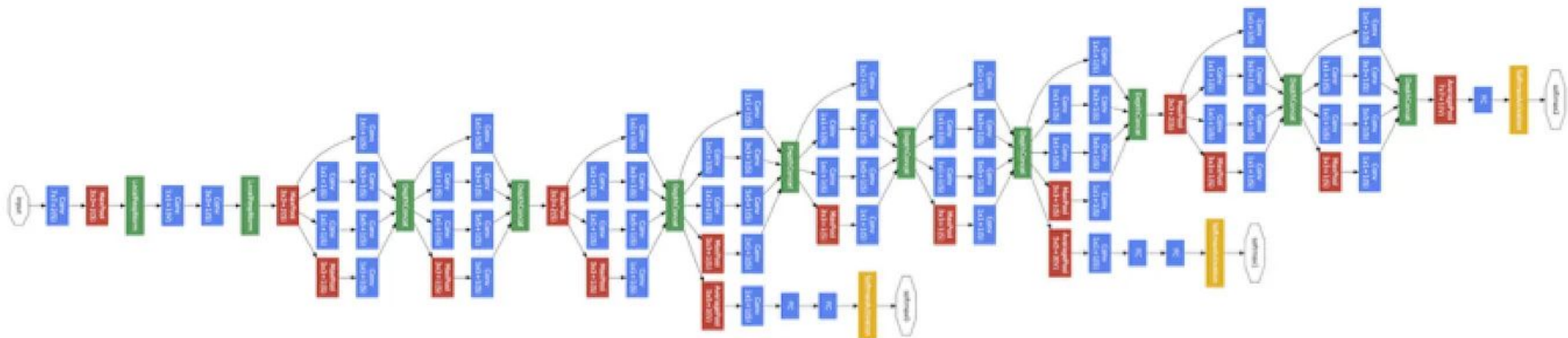
Deep Layers - higher level features such as dogs faces or flowers



GoogLeNet / Inception (2014)

- The winner of the ILSVRC 2014 competition was GoogLeNet (a.k.a. Inception V1) from Google.
 - It achieved a top-5 error rate of 6.67%!
 - This was very close to human level performance which the organisers of the challenge were now forced to evaluate.
-

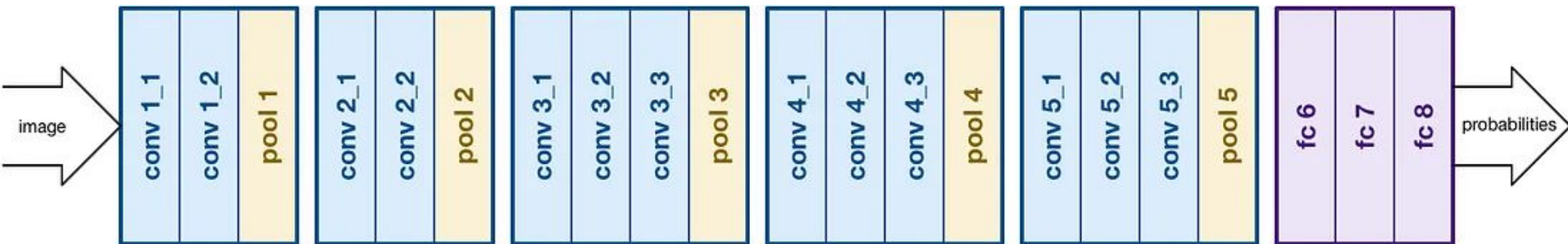
- The network used a CNN inspired by LeNet but implemented a novel element which is dubbed an inception module.
- It used batch normalization, image distortions and RMSprop.
- This module is based on several very small convolutions in order to drastically reduce the number of parameters. Their architecture consisted of a 22 layer deep CNN but reduced the number of parameters from 60 million (AlexNet) to 4 million.



Convolution
Pooling
Softmax
Other

VGGNet (2014)

- The runner-up at the ILSVRC 2014 competition is dubbed VGGNet by the community and was developed by Simonyan and Zisserman.
- VGGNet consists of 16 convolutional layers and is very appealing because of its very uniform architecture.
- Similar to AlexNet, only 3x3 convolutions, but lots of filters.
- Trained on 4 GPUs for 2–3 weeks. It is currently the most preferred choice in the community for extracting features from images.
- The weight configuration of the VGGNet is publicly available and has been used in many other applications and challenges as a baseline feature extractor. However, VGGNet consists of 138 million parameters, which can be a bit challenging to handle.



cont...

Idea:

- stack the convolutional layers with increasing filter sizes
- 3 x 3 filter size stride of 1
- MaxPooling 2 x 2 stride of 2
- **VGG Block** • Dropout, MaxPooling, ReLu
- On a system equipped with four NVIDIA Titan Black GPUs, training a single net took 2-3 weeks depending on the architecture

6 VGGNet Architectures

architectures															Number of Parameters (millions)	Top-5 Error Rate (%)					
Image	Conv3-64	Max pool	Conv3-128	Max pool	Conv3-256	Conv3-256	Max pool	Conv3-512	Conv3-512	Max pool	Conv3-512	Conv3-512	Max pool	FC-4096	FC-4096	FC-1000	Soft-max	133	10.4		
VGG-11																					
Image	Conv3-64	LRN	Max pool	Conv3-128	Max pool	Conv3-256	Conv3-256	Max pool	Conv3-512	Conv3-512	Max pool	Conv3-512	Conv3-512	Max pool	FC-4096	FC-4096	FC-1000	Soft-max	133	10.5	
VGG-11 (LRN)																					
Image	Conv3-64	Conv3-64	Max pool	Conv3-128	Conv3-128	Max pool	Conv3-256	Conv3-256	Max pool	Conv3-512	Conv3-512	Max pool	Conv3-512	Conv3-512	Max pool	FC-4096	FC-4096	FC-1000	Soft-max	133	9.9
VGG-13																					
Image	Conv3-64	Conv3-64	Max pool	Conv3-128	Conv3-128	Max pool	Conv3-256	Conv3-256	Conv1-256	Max pool	Conv3-512	Conv3-512	Conv1-512	Max pool	FC-4096	FC-4096	FC-1000	Soft-max	134	9.4	
VGG-16 (Conv1)																					
Image	Conv3-64	Conv3-64	Max pool	Conv3-128	Conv3-128	Max pool	Conv3-256	Conv3-256	Conv3-256	Max pool	Conv3-512	Conv3-512	Conv3-512	Max pool	FC-4096	FC-4096	FC-1000	Soft-max	138	8.8	
VGG-16																					
Image	Conv3-64	Conv3-64	Max pool	Conv3-128	Conv3-128	Max pool	Conv3-256	Conv3-256	Conv3-256	Max pool	Conv3-512	Conv3-512	Conv3-512	Max pool	FC-4096	FC-4096	FC-1000	Soft-max	144	9.0	
VGG-19																					

VGG-11

VGG-11 (LRN)

VGG-13

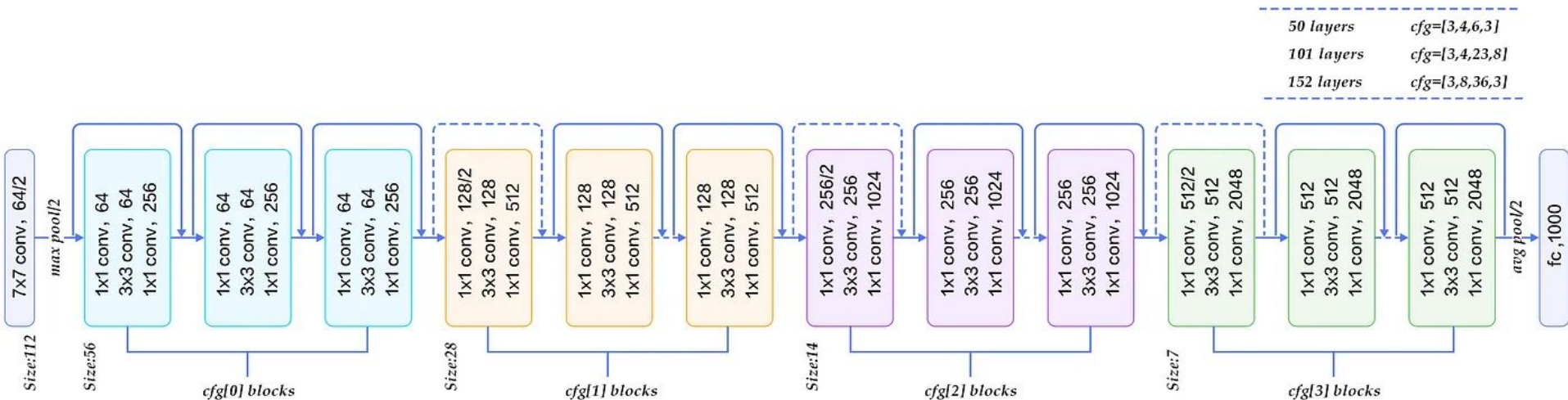
VGG-16 (Conv1)

VGG-16

VGG-19

ResNet (2015)

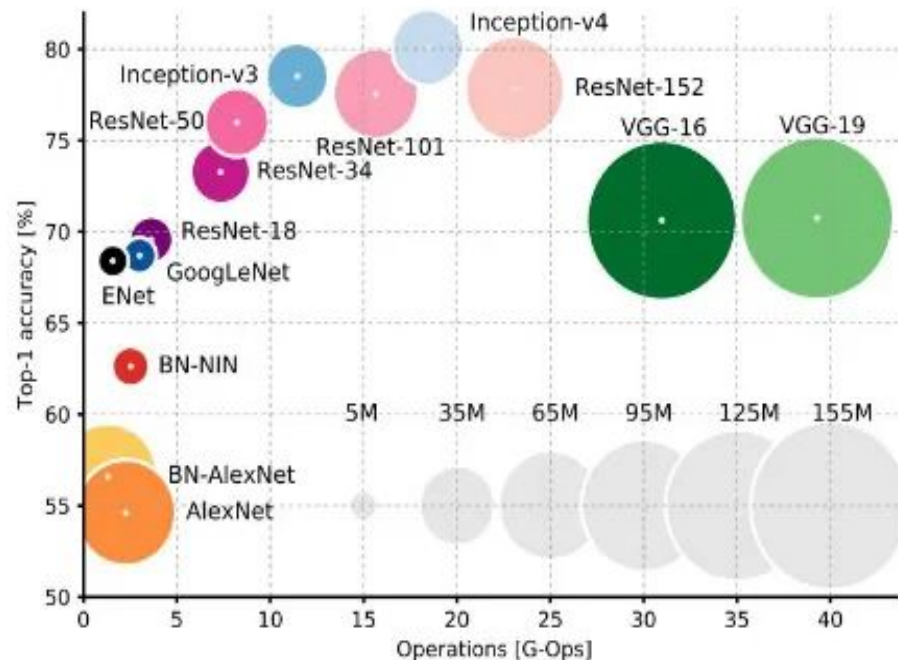
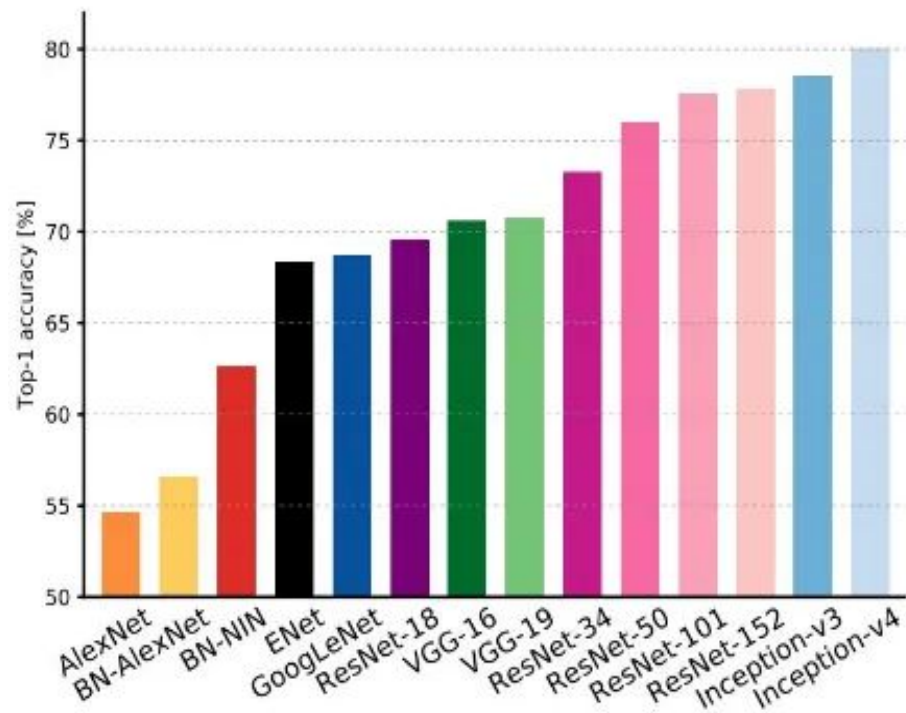
- At the ILSVRC 2015, the so-called Residual Neural Network (ResNet) by Kaiming He et al introduced a novel architecture with “skip connections” and features heavy batch normalization.
- Such skip connections are also known as gated units or gated recurrent units and have a strong similarity to recent successful elements applied in RNNs.
- This technique, they were able to train a NN with 152 layers while still having lower complexity than VGGNet.
- It achieves a top-5 error rate of 3.57% which beats human-level performance on this dataset.



Points to remember

- **AlexNet** has parallel two CNN line trained on two GPUs with cross-connections
- **GoogleNet** has inception modules
- **ResNet** has residual connections

Analysis of Deep Neural Network Models



An Analysis of Deep Neural Network Models for Practical Applications, 2017.

Summary

Year	CNN	Developed by	Place	Top-5 error rate	No. of parameters
1998	LeNet(8)	Yann LeCun et al			60 thousand
2012	AlexNet(7)	Alex Krizhevsky, Geoffrey Hinton, Ilya Sutskever	1st	15.3%	60 million
2013	ZFNet()	Matthew Zeiler and Rob Fergus	1st	14.8%	
2014	GoogLeNet(19)	Google	1st	6.67%	4 million
2014	VGG Net(16)	Simonyan, Zisserman	2nd	7.3%	138 million
2015	ResNet(152)	Kaiming He	1st	3.6%	

CovNet Architectures

- LeNet (1990s)
- AlexNet (2012)
- ZF Net (2013)
- VGGNet (2014)
- GoogLeNet (2014)
- ResNets (2015)
- DenseNet (2017)

DenseNet (2017)

- Connects all layers directly with each other
- Handle vanishing gradients problem
- Addition vs. concatenation (ResNet vs. DenseNet)
- Dense blocks

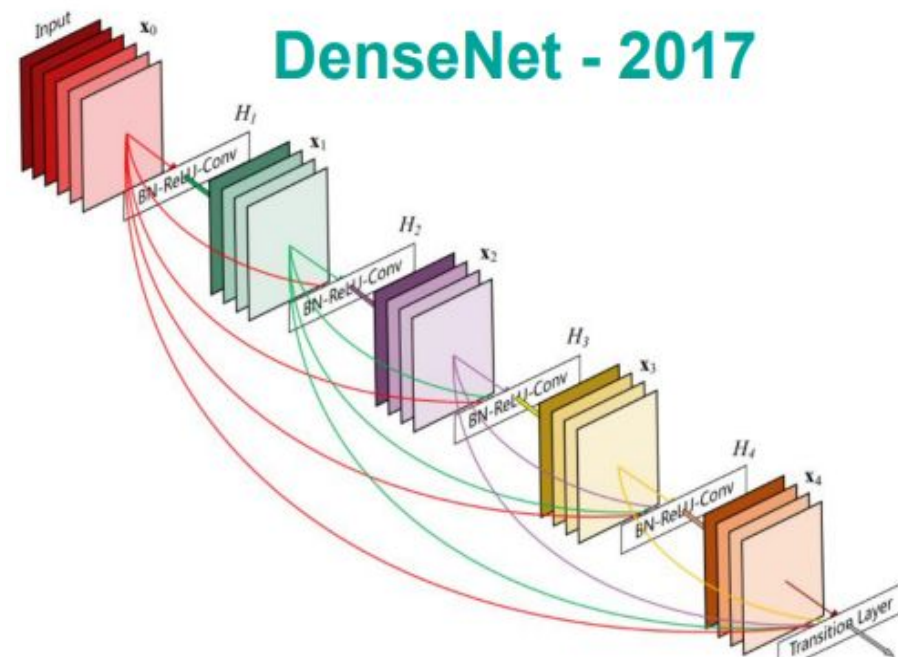
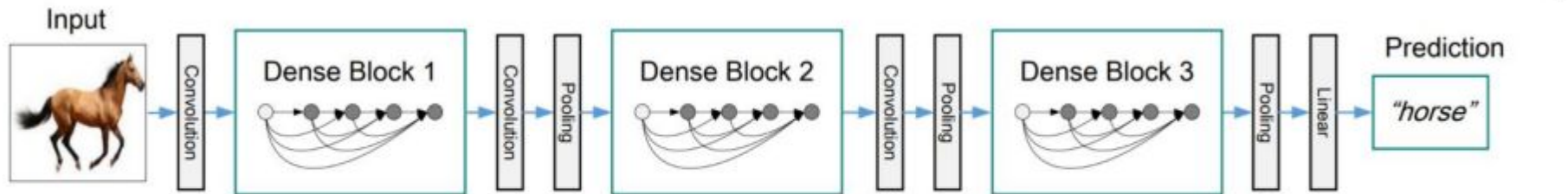


Figure 2: A deep DenseNet with three dense blocks. The layers between two adjacent blocks are referred to as transition layers and change feature-map sizes via convolution and pooling.

Refer DenseNet ppt for more information..

Thank You